

Niloy Saha

Distributed Systems | 5G / Cloud-Native Infrastructure | Network Telemetry & Automation

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Profile

Ph.D. researcher in Computer Science at the University of Waterloo specializing in scalable and autonomous networked systems. Designed and deployed cloud-native monitoring and control mechanisms for distributed infrastructures, with a focus on observability, anomaly detection, and resource-efficient operation. Validated in full-stack 5G deployments, these systems reduce service-level agreement violations by 2.3x and telemetry overhead by 10x while preserving detection accuracy under fixed resource budgets. Published in leading networking venues (ACM SIGMETRICS, IEEE TNSM) and collaborated with industry partners on the deployment of private 5G network.

Experience

University of Waterloo

Sep 2020 – Present

Ph.D. Researcher | Distributed Systems

- Architected and implemented a scalable cross-layer monitoring platform for 5G network slices, delivering predictable overhead scaling and sub-3 ms ingestion latency [\[Paper\]](#).
- Developed programmable user-plane telemetry in P4 on Intel Tofino switches, reducing monitoring overhead by **10x** compared to packet sampling approaches [\[Paper\]](#).
- Designed resource-aware telemetry allocation mechanisms to dynamically adjust monitoring fidelity under fixed compute and bandwidth budgets, and reduce service-level agreement (SLA) violations by **2.3x** [\[Paper\]](#).
- Deployed and operated a 5G testbed integrating containerized core functions, telemetry exporters, CI/CD pipelines, and observability infrastructure [\[GitHub\]](#).
- Collaborated with Rogers and Expeto on the Waterloo campus private 5G deployment and participated on technology transfer workshops on slice orchestration and monitoring [\[Workshop\]](#).

Indian Institute of Technology Kharagpur

2016 – 2019

Graduate Researcher | Software-defined Networks

- Developed SDN-based QoS mechanisms reducing latency by **35%** and packet loss by **12%** [\[Paper\]](#).
- Designed fog-computing task offloading algorithms under dynamic workloads [\[Paper\]](#).
- Led deployment of a cloud-assisted telemedicine communication system.

Education

Ph.D., Computer Science

2020 – 2026

University of Waterloo, Canada

Successfully defended: June 2026

GPA: 95.2/100

M.S., Computer Science

2016 – 2019

Indian Institute of Technology Kharagpur, India

GPA: 9.51/10

B.Tech., Electronics & Communication Engineering

2012 – 2016

Institute of Engineering and Management, India

GPA: 9.07/10

Technical Expertise

Cloud-Native: Kubernetes, Docker, Helm, Linux, CI/CD

Networking & Systems: NS-3, ONOS, POX, Intel Tofino (P4), Open vSwitch

Observability & Reliability: Prometheus, Grafana, telemetry pipelines, SLA monitoring

Machine Learning: PyTorch, scikit-learn, Reinforcement Learning (RL)

Programming: Python, C/C++, Bash

Selected Publications

ACM SIGMETRICS 2026 - Dynamic SLA-aware Network Slice Monitoring

IEEE/IFIP NOMS 2026 - Sketch-Based Telemetry for QoS Anomaly Detection

Best Paper Award

IEEE TNSM 2025 - Monitoring Architecture for 5G Network Slices

IEEE JSAC 2024 - Flexible RAN Slicing in Open RAN (co-author)

Full list: [Google Scholar](#)

Awards

Best Paper Award, IEEE/IFIP NOMS 2026

Mitacs Accelerate Fellowship (2021–2024)

International Doctoral Student Award (2020–2024)

IEEE Communications Society Student Grant, ICC 2020

Professional Activities

Web Chair, IEEE/IFIP NOMS 2025

Reviewer: IEEE/ACM ToN, TMC, JSAC; major networking conferences

Last updated: June 22, 2026